

Supplementary Material

Landscape composition and configuration as drivers of the Effective Habitat Area of threatened mammals in the Atlantic Forest of Rio de Janeiro State

This document provides supporting information for the analyses presented in the main manuscript. Section numbers follow the structure of the manuscript. Tables labelled S1 to S10 are those directly referenced in the main text; Tables S11 to S15 provide additional results that complement the main findings. All landscape metrics were calculated for the Forest Vegetation class, the response variable Effective Habitat Area (EHA, in hectares) is the geodetic area of each Sampling Unit (SU) computed under the Albers Equal-Area Conic projection (standard parallels 7°S and 22°S; central meridian 54°W), and all randomisation-based tests used 9,999 permutations.

1. Materials and Methods

1.1 Aggregation of Land Use and Land Cover Classes (Section 2.4 of the manuscript)

The 22 terminal MapBiomas Collection 10.1 subclasses (Souza et al., 2020; MapBiomas, 2025) were aggregated into five ecologically interpretable categories prior to the multivariate analyses. The aggregation was guided by a two-stage spatial correlation analysis, in which 231 pairwise Pearson correlations were computed among all subclass binary rasters using a stratified sample of 100,000 cells. Of these five categories, four were retained as compositional predictors in the subsequent ecological analyses (Forest Vegetation, Farming, Herbaceous and Shrubby Vegetation, and Non-vegetated Areas). The Water category was excluded because the focal taxa are terrestrial medium and large-sized mammals, for which water bodies act as habitat boundaries rather than habitat per se.

Table S1. Description of the four Land Use and Land Cover (LULC) functional categories retained as compositional predictors, with the MapBiomas Collection 10.1 subclasses grouped within each.

Class name	Grouped subclasses	Description
Forest Vegetation	Forest Formation; Mangrove; Floodable Forest; Savanna Formation; Wooded Sandbank Vegetation	Natural areas predominantly covered by woody vegetation, including dense and open forest formations with continuous canopies, coastal mangrove ecosystems, periodically or permanently flooded forests, tree-dominated savanna formations, and coastal wooded restinga.
Farming	Pasture; Agriculture; Forest Plantation; Mosaic of Uses	Anthropogenic landscapes dedicated to agricultural and livestock activities, including planted and managed pastures, temporary and permanent crop fields, commercial forest plantations such as silviculture, and mixed mosaics where agricultural and pastoral land uses cannot be distinguished at the pixel resolution.
Herbaceous and	Wetland; Grassland; Hypersaline	Natural ecosystems dominated by non-woody, grass-

Shrubby Vegetation	Tidal Flat; Rocky Outcrop; Herbaceous Sandbank Vegetation	like, or low-stature vegetation. Includes natural grasslands, seasonally or permanently flooded herbaceous wetlands, hypersaline tidal flats (apicuns), sparsely vegetated rocky outcrops, and coastal herbaceous restinga formations.
Non-vegetated Areas	Beach, Dune and Sand Spot; Urban Area; Mining; Other Non-Vegetated Areas	Surfaces without persistent vegetation cover, including natural exposed substrates such as beaches, dunes and sand spots, and anthropogenic non-vegetated surfaces such as urban areas, mining sites, and bare ground.

1.2 Landscape Metrics: Definitions and Ecological Interpretation (Section 2.5 of the manuscript)

Figure S1. Pearson correlation matrix among all candidate landscape and socio-environmental predictors used in the collinearity diagnostics described in Section 2.6.1 of the manuscript. Pairwise coefficients with absolute values above 0.7 were flagged for further evaluation with Variance Inflation Factors (Table S3).

Table S2. Selected landscape metrics, units, calculation method and ecological interpretation. All metrics were calculated for the Forest Vegetation class.

Metric	Abbreviation	Unit	Ecological interpretation
Class Area	CA	Hectares (ha)	Total habitat amount (absolute extent of the focal class). Primary indicator of habitat loss at the landscape scale.
Percentage of Landscape	PLAND	Percentage (%)	Proportional cover of the focal class within the Effective Habitat Area. Standard measure of habitat amount, central to the Habitat Amount Hypothesis (Fahrig, 2013).
Number of Patches	NP	Count	Total number of discrete patches of the focal class within the EHA. Higher values denote stronger subdivision of habitat, and in fragmented landscapes function as a measure of stepping-stone availability.
Patch Density	PD	Number of patches per 100 ha (n/100 ha)	Number of patches standardised by landscape area, permitting comparison among SUs of different sizes. Captures fragmentation per se when sample units differ in extent.
Mean Patch Area	Mean_AREA	Hectares (ha)	Average patch size. Directly relates to the area requirements of forest-dependent species and to the magnitude of edge effects.
Largest Patch Index	LPI	Percentage (%)	Percentage of the total landscape area comprised by the single largest patch of the focal class. Indicator of dominance of a core forest remnant.
Edge Density	ED	Metres of edge per hectare (m/ha)	Total length of edge segments of the focal class per landscape unit area. Higher values denote increased exposure of forest interior to the surrounding matrix.
Mean Fractal Dimension Index	Mean_FRAC	Dimensionless (1-2)	Shape complexity of patch boundaries. Values approaching 2 indicate highly convoluted shapes and elevated edge-to-interior ratios.
Mean Euclidean Nearest-Neighbour Distance	Mean_ENN	Metres (m)	Average distance from each patch to its nearest neighbour of the same class. Inverse indicator of structural connectivity.
Mean Proximity Index	PROX_MN	Dimensionless	Aggregation and proximity of patches within a species-specific search radius (28,230 m, see below). Lower values indicate higher patch isolation.

1.2.1 Technical Notes on Metric Calculation

Mean Euclidean Nearest-Neighbour Distance (Mean_ENN). The ENN calculation considered the 30 m spatial resolution of the MapBiomas raster and an 8-neighbour rule for cell connectivity. Under this configuration, the minimum possible ENN value between two non-adjacent fragments is approximately 42.43 m, corresponding to the shortest diagonal connection. Values approaching this minimum indicate maximal proximity between habitat patches.

Mean Proximity Index (PROX_MN). To define an ecologically meaningful search radius for the PROX, we estimated the maximum dispersal distance (d) of each focal taxon following the isometric relationship of Bowman et al. (2002), $d = 40 \times \sqrt{A}$, where A is the home range area. Home range data were obtained from the HomeRange global database (Broekman et al., 2023). The mean of the resulting maximum dispersal distances across the focal taxa was 28,230 m, which was adopted as the PROX search radius. This radius reflects the functional connectivity scale relevant to medium and large-sized mammals crossing the fragmented matrix.

1.3 Collinearity Diagnostics (Section 2.6.1 of the manuscript)

Collinearity among landscape and socio-environmental predictors was evaluated using a two-stage procedure. First, the Pearson correlation matrix among all candidate predictors was inspected, and pairwise coefficients with absolute values above 0.7 were flagged (Figure S1). Second, the Variance Inflation Factor (VIF) was computed using the car package (Fox & Weisberg, 2019), with a threshold of 5.0 used to identify variables introducing multicollinearity into the multivariable models (Table S3). On the basis of these diagnostics, the Largest Patch Index (LPI) was excluded from the univariate models because of extreme collinearity with the Percentage of Landscape (PLAND). LPI was nevertheless retained in the multivariate ordination space, since the envfit procedure evaluates each vector independently rather than within a joint linear model and because LPI and PLAND capture structurally distinct ecological signatures (dominance of a single core remnant versus proportional cover).

Table S3. Variance Inflation Factor (VIF) values for all candidate landscape and socio-environmental predictors, sorted by descending VIF. A threshold of 5.0 was used to flag variables introducing multicollinearity into the multivariable models.

Predictor	VIF	Status
LPI	53.12	Excluded from univariate models (VIF > 5)
PLAND	48.79	Excluded from univariate models (VIF > 5)
PD	6.19	Excluded from univariate models (VIF > 5)
ED	4.23	Retained
Mean AREA (ha)	2.90	Retained
Mean ENN (m)	2.30	Retained
Mean FRAC	2.26	Retained
PROX_MN	1.96	Retained
NP	1.44	Retained
Mean Elevation (m)	1.43	Retained
Mean Pop. Density	1.26	Retained

Legend. VIF threshold: 5.0. Only LPI and PLAND exceeded the threshold. LPI was excluded from the univariate GLMM/GLM framework but retained in the multivariate envfit (Section 2.6.2 of the manuscript). All other predictors presented VIF values well below 5 and were retained.

1.4 Sample Sizes, Inclusion Rules and Model Diagnostics (Section 2.6.3 of the manuscript)

Groups containing fewer than five sampling units were excluded from the univariate GLMM/GLM framework to preserve statistical stability. Table S4 reports the per-group sample sizes and the inclusion status of each taxonomic and functional group. The PERMANOVA framework (Section 2.6.2 of the manuscript) retained all original categories, since it does not impose per-group sample-size thresholds. Residual diagnostics for all retained GLMM/GLM models were obtained from simulation-based tests implemented in the DHARMA package (Hartig, 2022) and are summarised in Table S5.

Table S4. Per-group sample sizes (number of Sampling Units, SU) and inclusion status in the univariate Generalized Linear (Mixed) Models. The total dataset comprises 65 SUs distributed across nine threatened taxa.

Category	Group	n (SU)	Status in univariate models
Genus	Puma concolor	12	Included
Genus	Alouatta guariba	12	Included
Genus	Brachyteles spp. (arachnoides + hypoxanthus)	11	Included
Genus	Mazama spp.	10	Included
Genus	Tayassu pecari	7	Included
Genus	Bradypus crinitus	7	Included
Genus	Leopardus wiedii	6	Included
Genus	Myrmecophaga tridactyla	4	Excluded (n < 5)
Diet	Frugivore/Folivore (Fr/Fo)	23	Included
Diet	Carnivore (Ca)	18	Included
Diet	Herbivore/Frugivore (Hb/Fr)	10	Included
Diet	Frugivore/Omnivore (Fr/On)	7	Included
Diet	Folivore (Fo)	5	Excluded (redundant with Bradypus at genus level)
Diet	Myrmecophagous (Myr)	4	Excluded (n < 5)
Locomotion	Terrestrial (Te)	33	Included
Locomotion	Arboreal (Ar)	28	Included
Locomotion	Scansorial (Sc)	6	Included

Legend. Genus-level sample sizes refer to the count of independent Sampling Units (EHA polygons) attributed to each genus. For *Brachyteles*, *B. arachnoides* and *B. hypoxanthus* contribute jointly to the genus-level model. The PERMANOVA framework retained all categories regardless of n.

Table S5. DHARMA simulation-based residual diagnostics for the retained GLMM and GLM models. Tests evaluate dispersion, uniformity of scaled residuals (Kolmogorov-Smirnov), and the proportion of outliers, all at $\alpha = 0.05$.

Category	Group	n	Dispersion p	Uniformity p (KS)	Outliers p	Diagnostic
Genus	Puma	12	0.758	0.6651	1	OK
Genus	Alouatta	11	0.132	0.3394	1	OK
Genus	Mazama	10	0.342	0.8977	1	OK
Genus	Brachyteles	10	0.73	0.6209	1	OK
Genus	Tayassu	7	0.916	0.7738	1	OK
Genus	Leopardus	6	0.85	0.9756	1	OK
Genus	Bradypus	5	0.772	0.5929	1	OK
Diet	Fr/On	7	0.916	0.7738	1	OK
Diet	Hb/Fr	10	0.342	0.8977	1	OK

Diet	Ca	18	0.56	0.8734	0.0354	WARNING
Diet	Fr/Fo	21	0.69	0.2266	1	OK
Locomotion	Ar	26	0.81	0.2254	1	OK
Locomotion	Sc	6	0.85	0.9756	1	OK
Locomotion	Te	33	0.114	0.4193	0.0639	OK

Legend. OK: assumptions met at $\alpha = 0.05$. WARNING: at least one test marginally violated (Carnivore diet had an outlier ratio slightly above the expected value). All other models satisfied the assumptions of dispersion, uniformity and outlier proportion under the Gamma error structure with log link.

2. Results

2.1 Landscape Structure in the Effective Habitat Areas (Section 4.1 of the manuscript)

The Non-metric Multidimensional Scaling (NMDS) ordination produced a stable two-dimensional representation of the landscape composition gradients across the Effective Habitat Areas of the focal taxa (stress = 0.089). Environmental vectors fitted onto the ordination space identified the configuration metrics most strongly aligned with the axes (Table S6). The homogeneity of multivariate dispersions was tested using the betadisper function, and the global PERMANOVA was complemented by post-hoc pairwise comparisons with False Discovery Rate (FDR) correction at the Genus, Species, Diet and Locomotion levels (Tables S7, S11, S12, S13).

Table S6. Environmental vectors fitted onto the NMDS ordination of landscape composition (envfit). R^2 and p-values reported by envfit are independent goodness-of-fit measures between each vector and the ordination scores and are therefore not partitioned variance components; they may sum to more than one across vectors and should be interpreted as marginal associations.

Metric	NMDS1	NMDS2	R^2	p-value
PLAND (%)	-0.084	-0.962	0.932	0.0001 ***
LPI (%)	-0.042	-0.925	0.857	0.0001 ***
PD	0.168	0.707	0.528	0.0001 ***
Mean PROX	-0.179	-0.621	0.418	0.0001 ***
ED	-0.124	0.359	0.144	0.0081 **
Mean FRAC	-0.057	0.127	0.019	0.548
Mean ENN (m)	-0.098	0.045	0.012	0.7038
Mean AREA (ha)	0.058	-0.080	0.010	0.7358

Legend. Significance codes: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. p-values estimated from 9,999 permutations.

Table S7. Pairwise PERMANOVA at the Genus level (Bray-Curtis dissimilarity, 9,999 permutations) with False Discovery Rate (FDR) correction.

Pair	pseudo-F	R^2	p-value	p-adjusted (FDR)
Alouatta vs Brachyteles	7.029	0.270	0.0011	0.0154 *
Alouatta vs Bradypus	0.783	0.053	0.5237	0.6094
Alouatta vs Leopardus	0.471	0.030	0.7418	0.7693
Alouatta vs Myrmecophaga	0.750	0.054	0.5441	0.6094
Alouatta vs Mazama	2.085	0.099	0.1189	0.2245
Alouatta vs Puma	1.997	0.087	0.1232	0.2245
Alouatta vs Tayassu	2.998	0.158	0.0306	0.1224
Brachyteles vs Bradypus	7.329	0.360	0.0031	0.0289 *
Brachyteles vs Leopardus	5.954	0.298	0.0050	0.0350 *

Brachyteles vs Myrmecophaga	3.248	0.213	0.0301	0.1224
Brachyteles vs Mazama	2.019	0.101	0.1307	0.2245
Brachyteles vs Puma	7.836	0.281	0.0008	0.0154 *
Brachyteles vs Tayassu	2.290	0.132	0.0985	0.2122
Bradypus vs Leopardus	0.708	0.073	0.6052	0.6518
Bradypus vs Myrmecophaga	3.155	0.311	0.0732	0.1945
Bradypus vs Mazama	2.846	0.180	0.0654	0.1945
Bradypus vs Puma	0.284	0.019	0.8193	0.8193
Bradypus vs Tayassu	2.944	0.228	0.0647	0.1945
Leopardus vs Myrmecophaga	1.258	0.136	0.2880	0.3840
Leopardus vs Mazama	2.303	0.141	0.0845	0.1972
Leopardus vs Puma	1.231	0.071	0.3190	0.4060
Leopardus vs Tayassu	2.490	0.185	0.0764	0.1945
Myrmecophaga vs Mazama	0.966	0.074	0.3891	0.4737
Myrmecophaga vs Puma	1.972	0.123	0.1363	0.2245
Myrmecophaga vs Tayassu	1.419	0.136	0.2396	0.3354
Mazama vs Puma	3.665	0.155	0.0223	0.1224
Mazama vs Tayassu	1.673	0.100	0.1944	0.2865
Puma vs Tayassu	1.899	0.101	0.1566	0.2436

Legend. Significance code: * p -adjusted < 0.05. Pairs involving *Brachyteles* versus *Alouatta*, *Bradypus*, *Leopardus* and *Puma* were the only contrasts that remained significant after FDR correction, indicating that the southern muriqui is the main driver of taxonomic divergence in landscape occupation.

Table S8. Variation in forest cover (PLAND) within the Effective Habitat Areas of the focal taxa.

Species	PLAND minimum (%)	PLAND maximum (%)	PLAND mean (%)
<i>Alouatta guariba</i>	4.51	87.24	34.52
<i>Brachyteles arachnoides</i>	20.38	98.46	73.57
<i>Bradypus crinitus</i>	33.22	80.16	53.96
<i>Brachyteles hypoxanthus</i>	25.94	94.19	60.06
<i>Leopardus wiedii</i>	4.23	90.93	52.62
<i>Myrmecophaga tridactyla</i>	27.18	54.86	37.00
<i>Mazama spp.</i>	10.81	95.53	41.02
<i>Puma concolor</i>	28.06	97.77	59.64
<i>Tayassu pecari</i>	30.26	97.73	63.21

Legend. PLAND values quantify the proportion of forest cover within each Effective Habitat Area. *Brachyteles arachnoides* occupied EHAs with the highest mean forest cover, whereas *Alouatta guariba* and *Leopardus wiedii* persisted in landscapes with forest cover as low as 4.23%.

2.2 Effects of Landscape and Socio-Environmental Predictors on the EHA (Section 4.2 of the manuscript)

Generalized Linear Mixed Models (GLMMs) and reduced Generalized Linear Models (GLMs, for single-species groups or non-convergent cases) were fitted under an information-theoretic framework. Tables S9 and S10 present, respectively, the full set of competitive models ($\Delta\text{AICc} < 2.0$) and the relative importance of each predictor measured as the sum of Akaike weights ($\sum w_i$) across all competitive models in which the variable appeared. Table S15 reports the three best-supported models per group for additional transparency.

Table S9. Full competitive model set ($\Delta AICc < 2.0$) for each genus and functional group. The response variable is the Effective Habitat Area (EHA, in hectares). All models used a Gamma error distribution with a log link function; species identity was included as a random intercept (1 | SPECIES) when more than one species was present in the group, otherwise a fixed-effects GLM was fitted.

Category	Group	Rank	Selected variables	df	LogLik	AICc	$\Delta AICc$	Akaike weight
Genus	Puma	1	NP_Forest + PD_Forest	4	-129.153	272.021	0.000	0.510
Genus	Puma	2	NP_Forest + PROX_MN_Forest	4	-129.193	272.100	0.079	0.490
Genus	Alouatta	1	Mean_Pop_Density	3	-130.969	271.367	0.000	0.500
Genus	Alouatta	2	~ Null Model ~	2	-133.488	272.475	1.108	0.287
Genus	Alouatta	3	NP_Forest	3	-131.822	273.072	1.705	0.213
Genus	Mazama	1	NP_Forest + PD_Forest	4	-93.584	203.167	0.000	1.000
Genus	Brachyteles	1	NP_Forest + PLAND_Forest	4	-92.209	200.418	0.000	0.539
Genus	Brachyteles	2	NP_Forest + PD_Forest	4	-92.364	200.728	0.311	0.461
Genus	Tayassu	1	Mean_Pop_Density	3	-69.281	152.561	0.000	0.636
Genus	Tayassu	2	~ Null Model ~	2	-73.340	153.680	1.119	0.364
Genus	Leopardus	1	~ Null Model ~	2	-73.487	154.974	0.000	1.000
Genus	Bradypus	1	~ Null Model ~	2	-57.637	125.275	0.000	0.592
Genus	Bradypus	2	NP_Forest	3	-48.009	126.019	0.744	0.408
Diet	Fr/On	1	Mean_Pop_Density	3	-69.281	152.561	0.000	0.636
Diet	Fr/On	2	~ Null Model ~	2	-73.340	153.680	1.119	0.364
Diet	Hb/Fr	1	NP_Forest + PD_Forest	4	-93.584	203.167	0.000	1.000
Diet	Ca	1	NP_Forest + PROX_MN_Forest	5	-201.326	417.652	0.000	1.000
Diet	Fr/Fo	1	NP_Forest	4	-240.393	491.285	0.000	1.000
Locomotion	Ar	1	NP_Forest	4	-299.185	608.274	0.000	0.447
Locomotion	Ar	2	NP_Forest + PROX_MN_Forest	5	-298.459	609.918	1.644	0.196
Locomotion	Ar	3	NP_Forest + PD_Forest	5	-298.546	610.091	1.817	0.180
Locomotion	Ar	4	ED_Forest + NP_Forest	5	-298.565	610.130	1.856	0.177
Locomotion	Sc	1	~ Null Model ~	2	-73.487	154.974	0.000	1.000
Locomotion	Te	1	Mean_ENN_Forest_m + NP_Forest + PD_Forest	6	-343.387	702.004	0.000	1.000

Legend. Variables: NP_Forest = Number of Patches; PD_Forest = Patch Density; PLAND_Forest = Percentage of Forest Landscape; PROX_MN_Forest = Mean Proximity Index; Mean_ENN_Forest_m = Mean Euclidean Nearest-Neighbour Distance; ED_Forest = Edge Density; Mean_Pop_Density = Human Population Density.

Table S10. Relative importance (sum of Akaike weights, Σw_i) of each predictor for each genus, diet and locomotion category. Predictors are listed in descending order of Σw_i within each group; only predictors with $\Sigma w_i > 0.01$ are shown for clarity. Full numerical values are available in the project repository.

Category	Group	Predictor	Σw_i
Genus	Puma	NP	0.965
		PROX	0.386
		PD	0.384
		PLAND (%)	0.110
		FRAC	0.040
		ED	0.015
		AREA (ha)	0.012
Genus	Alouatta	Pop. Density	0.429

		NP	0.181
		PLAND (%)	0.125
		ED	0.085
		ENN (m)	0.084
		FRAC	0.073
		Elevation (m)	0.069
		AREA (ha)	0.064
		PD	0.064
		PROX	0.058
Genus	Mazama	NP	1.000
		PD	0.999
Genus	Brachyteles	NP	0.950
		PLAND (%)	0.471
		PD	0.403
		PROX	0.123
		ED	0.036
Genus	Tayassu	Pop. Density	0.490
		NP	0.116
		PD	0.048
		AREA (ha)	0.038
		Elevation (m)	0.024
		ED	0.017
		PROX	0.015
		PLAND (%)	0.013
		ENN (m)	0.011
Genus	Bradypus	NP	0.408
Genus	Leopardus	NP	0.026
		Elevation (m)	0.024
		ENN (m)	0.016
		FRAC	0.012
		Pop. Density	0.010
Diet	Fr/Fo	NP	0.666
		ED	0.198
		PLAND (%)	0.130
		ENN (m)	0.100
		PROX	0.100
		Pop. Density	0.085
		PD	0.084
		FRAC	0.078
		AREA (ha)	0.073
		Elevation (m)	0.071
Diet	Ca	NP	0.999
		PROX	0.900
		PD	0.073
		ED	0.015
Diet	Hb/Fr	NP	1.000
		PD	0.999
Diet	Fr/On	Pop. Density	0.490
		NP	0.116
		PD	0.048
		AREA (ha)	0.038

Locomotion	Ar	Elevation (m)	0.024
		ED	0.017
		PROX	0.015
		PLAND (%)	0.013
		ENN (m)	0.011
		NP	0.848
		ED	0.162
		PROX	0.112
		PD	0.097
		ENN (m)	0.091
Locomotion	Te	Elevation (m)	0.086
		PLAND (%)	0.083
		FRAC	0.078
		Pop. Density	0.069
		AREA (ha)	0.063
		NP	1.000
		PD	0.999
		ENN (m)	0.585
		FRAC	0.068
		AREA (ha)	0.054
Locomotion	Sc	Elevation (m)	0.051
		Pop. Density	0.047
		NP	0.026
		Elevation (m)	0.024
		ENN (m)	0.016
		FRAC	0.012
		Pop. Density	0.010

Legend. Σw_i values approaching 1 denote consistently supported predictors across the competitive model set. NP = Number of Patches; PD = Patch Density; PLAND = Percentage of Forest Landscape; PROX = Mean Proximity Index; ENN = Mean Euclidean Nearest-Neighbour Distance; ED = Edge Density; FRAC = Mean Fractal Dimension; AREA = Mean Patch Area; Pop. Density = Human Population Density.

3. Additional Supporting Tables

The tables below complement those directly cited in the main manuscript. Table S11 reports the full betadisper results that support the assumption check of the global PERMANOVA presented as Table 1 in the manuscript. Tables S12 to S14 report pairwise PERMANOVA contrasts at the Species, Diet and Locomotion levels. Table S15 provides the top three competing models per group, ranked by ΔAIC_c .

Table S11. Homogeneity of multivariate dispersions test (betadisper) for each grouping variable (Bray-Curtis dissimilarity). Heterogeneity of dispersions does not invalidate PERMANOVA but indicates that group differences may partly reflect variance heterogeneity rather than location shifts.

Grouping	Source	Df	Sum Sq	Mean Sq	F	Pr(>F)
Genus	Groups	7	0.171	0.0244	1.844	0.0964
Genus	Residuals	57	0.755	0.0132	NA	NA
Species	Groups	8	0.179	0.0223	1.764	0.1040
Species	Residuals	56	0.709	0.0127	NA	NA
Diet	Groups	5	0.229	0.0458	4.163	0.0026

Diet	Residuals	59	0.649	0.0110	NA	NA
Locomotion	Groups	2	0.063	0.0315	2.047	0.1378
Locomotion	Residuals	62	0.954	0.0154	NA	NA

Legend. Dispersions were homogeneous for Genus, Species and Locomotion ($p > 0.05$). For Diet, dispersions were significantly heterogeneous ($F = 4.16$, $p = 0.003$), so the Diet-level PERMANOVA results in the main manuscript are reported with this caveat and the Diet-level pairwise tests in Table S13 should be regarded as exploratory.

Table S12. Pairwise PERMANOVA at the Species level (Bray-Curtis dissimilarity, 9,999 permutations) with False Discovery Rate (FDR) correction. No species pair remained significant after FDR correction.

Pair	pseudo-F	R²	p-value	p-adjusted (FDR)
A. guariba vs B. arachnoides	5.570	0.247	0.0030	0.0864
A. guariba vs B. crinitus	0.783	0.053	0.5237	0.5892
A. guariba vs B. hypoxantus	3.441	0.238	0.0534	0.1618
A. guariba vs L. wiedii	0.471	0.030	0.7418	0.7630
A. guariba vs M. tridactyla	0.750	0.054	0.5441	0.5936
A. guariba vs Mazama	2.085	0.099	0.1189	0.2218
A. guariba vs P. concolor	1.997	0.087	0.1232	0.2218
A. guariba vs T. pecari	2.998	0.158	0.0306	0.1377
B. arachnoides vs B. crinitus	6.386	0.367	0.0092	0.0864
B. arachnoides vs B. hypoxantus	1.234	0.134	0.2626	0.3521
B. arachnoides vs L. wiedii	4.808	0.286	0.0120	0.0864
B. arachnoides vs M. tridactyla	2.952	0.228	0.0556	0.1618
B. arachnoides vs Mazama	1.686	0.095	0.1882	0.2916
B. arachnoides vs P. concolor	5.627	0.238	0.0063	0.0864
B. arachnoides vs T. pecari	1.410	0.098	0.2641	0.3521
B. crinitus vs B. hypoxantus	7.837	0.611	0.0476	0.1618
B. crinitus vs L. wiedii	0.708	0.073	0.6052	0.6408
B. crinitus vs M. tridactyla	3.155	0.311	0.0732	0.1618
B. crinitus vs Mazama	2.846	0.180	0.0654	0.1618
B. crinitus vs P. concolor	0.284	0.019	0.8193	0.8193
B. crinitus vs T. pecari	2.944	0.228	0.0647	0.1618
B. hypoxantus vs L. wiedii	3.220	0.349	0.0717	0.1618
B. hypoxantus vs M. tridactyla	3.832	0.489	0.0667	0.1618
B. hypoxantus vs Mazama	1.377	0.121	0.2865	0.3575
B. hypoxantus vs P. concolor	6.592	0.355	0.0111	0.0864
B. hypoxantus vs T. pecari	3.654	0.343	0.0297	0.1377
L. wiedii vs M. tridactyla	1.258	0.136	0.2880	0.3575
L. wiedii vs Mazama	2.303	0.141	0.0845	0.1690
L. wiedii vs P. concolor	1.231	0.071	0.3190	0.3828
L. wiedii vs T. pecari	2.490	0.185	0.0764	0.1618
M. tridactyla vs Mazama	0.966	0.074	0.3891	0.4519
M. tridactyla vs P. concolor	1.972	0.123	0.1363	0.2337
M. tridactyla vs T. pecari	1.419	0.136	0.2396	0.3450
Mazama vs P. concolor	3.665	0.155	0.0223	0.1338
Mazama vs T. pecari	1.673	0.100	0.1944	0.2916
P. concolor vs T. pecari	1.899	0.101	0.1566	0.2563

Legend. Significance code: * p -adjusted < 0.05 . The absence of significant species-level contrasts after FDR correction is consistent with the genus-level pattern, where *Brachyteles* emerges as the main source of taxonomic divergence in landscape composition.

Table S13. Pairwise PERMANOVA at the Diet level (Bray-Curtis dissimilarity, 9,999 permutations) with FDR correction. These contrasts should be regarded as exploratory in view of the heterogeneity of multivariate dispersions detected for Diet (see Table S11).

Pair	pseudo-F	R ²	p-value	p-adjusted (FDR)
Fr/Fo vs Fo	1.573	0.061	0.1978	0.3594
Fr/Fo vs Ca	2.562	0.065	0.0662	0.2196
Fr/Fo vs Myr	0.451	0.019	0.7180	0.7693
Fr/Fo vs Hb/Fr	0.551	0.019	0.6487	0.7485
Fr/Fo vs Fr/On	0.821	0.031	0.4806	0.6008
Fo vs Ca	0.269	0.013	0.8497	0.8497
Fo vs Myr	3.155	0.311	0.0732	0.2196
Fo vs Hb/Fr	2.846	0.180	0.0654	0.2196
Fo vs Fr/On	2.944	0.228	0.0647	0.2196
Ca vs Myr	1.473	0.069	0.2352	0.3594
Ca vs Hb/Fr	3.922	0.131	0.0140	0.2100
Ca vs Fr/On	2.207	0.088	0.1081	0.2702
Myr vs Hb/Fr	0.966	0.074	0.3891	0.5306
Myr vs Fr/On	1.419	0.136	0.2396	0.3594
Hb/Fr vs Fr/On	1.673	0.100	0.1944	0.3594

Legend. Diet categories: Fr/Fo = Frugivore/Folivore; Fo = Folivore; Ca = Carnivore; Myr = Myrmecophagous; Hb/Fr = Herbivore/Frugivore; Fr/On = Frugivore/Omnivore.

Table S14. Pairwise PERMANOVA at the Locomotion level (Bray-Curtis dissimilarity, 9,999 permutations) with FDR correction.

Pair	pseudo-F	R ²	p-value	p-adjusted (FDR)
Ar vs Sc	1.527	0.048	0.1976	0.2964
Ar vs Te	0.307	0.005	0.8145	0.8145
Sc vs Te	2.349	0.060	0.0731	0.2193

Legend. Locomotion categories: Ar = Arboreal; Sc = Scansorial; Te = Terrestrial. No pairwise contrast was significant after FDR correction, consistent with the non-significant global PERMANOVA for Locomotion (Table 1 of the manuscript).

Table S15. Top three best-supported models per genus, ranked by ΔAICc . The full competitive set ($\Delta\text{AICc} < 2.0$) is reported in Table S9.

Genus	Rank	ΔAICc	Akaike weight	df	LogLik	Selected variables
Puma	1	0.000	0.446	4	-129.153	NP_Forest + PD_Forest
Puma	2	0.079	0.428	4	-129.193	NP_Forest + PROX_MN_Forest
Puma	3	2.528	0.126	4	-130.418	NP_Forest + PLAND_Forest
Alouatta	1	0.000	0.500	3	-130.969	Mean_Pop_Density
Alouatta	2	1.108	0.287	2	-133.488	~ Null Model ~
Alouatta	3	1.705	0.213	3	-131.822	NP_Forest
Mazama	1	0.000	0.999	4	-93.584	NP_Forest + PD_Forest
Mazama	2	13.632	0.001	4	-100.400	NP_Forest + PLAND_Forest
Mazama	3	22.691	0.000	4	-104.929	NP_Forest + PROX_MN_Forest
Brachyteles	1	0.000	0.496	4	-92.209	NP_Forest + PLAND_Forest
Brachyteles	2	0.311	0.425	4	-92.364	NP_Forest + PD_Forest
Brachyteles	3	3.676	0.079	4	-94.047	NP_Forest + PROX_MN_Forest
Tayassu	1	0.000	0.577	3	-69.281	Mean_Pop_Density
Tayassu	2	1.119	0.330	2	-73.340	~ Null Model ~
Tayassu	3	3.642	0.093	3	-71.102	NP_Forest
Leopardus	1	0.000	0.946	2	-73.487	~ Null Model ~

Leopardus	2	7.002	0.029	3	-71.988	NP_Forest
Leopardus	3	7.226	0.026	3	-72.100	Mean_Elevation_m
Bradypus	1	0.000	0.592	2	-57.637	~ Null Model ~
Bradypus	2	0.744	0.408	3	-48.009	NP_Forest
Bradypus	3	14.761	0.000	3	-55.018	PD_Forest

Legend. The Akaike weight reported here is computed within the top-3 set and may differ from the weight reported in Table S9, which is computed within the competitive set ($\Delta AICc < 2.0$).

4. References

The references listed below correspond to the works cited in this Supplementary Material. All references are also available in the main manuscript reference list, with the exception of those marked with an asterisk, which are cited only in the supplementary content.

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Asterisk () indicates a reference cited only in the Supplementary Material and not in the main manuscript reference list. We recommend that these two references be added to the main manuscript reference list, since they are also cited there (Section 2.6.1 for Fox & Weisberg, 2019, and Section 2.6.3 for Burnham & Anderson, 2002), but currently appear only here.*